
Statistical Inference

B. Statistical Data Science 2nd Year Indian Statistical Institute

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Exercise Series 13

Exercise 1. Let $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Bernoulli}(p_1)$ and $Y_1, \dots, Y_m \stackrel{\text{i.i.d.}}{\sim} \text{Bernoulli}(p_2)$ be two independent samples. We want to test $\mathcal{H}_0 : p_1 = p_2$ against $\mathcal{H}_a : p_1 \neq p_2$.

- (a) Formulate the Wald's test, Rao's score test and likelihood ratio test for this problem. Clearly specify the test statistics and the cut-off values.
- (b) How would you test $\mathcal{H}_0 : p_1 = p_2$ against $\mathcal{H}_a : p_1 > p_2$?

Exercise 2. Let $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Normal}(\mu_1, \sigma^2)$ and $Y_1, \dots, Y_m \stackrel{\text{i.i.d.}}{\sim} \text{Normal}(\mu_2, \sigma^2)$ be two independent samples, where $\mu_1 \in \mathbb{R}$, $\mu_2 \in \mathbb{R}$ and $\sigma^2 > 0$ are unknown parameters. We want to test $\mathcal{H}_0 : \mu_1 = \mu_2$ against $\mathcal{H}_a : \mu_1 \neq \mu_2$.

- (a) Formulate the Wald's test, Rao's score test and likelihood ratio test for this problem.
- (b) Formulate a test for $\mathcal{H}_0 : \mu_1 = \mu_2$ against $\mathcal{H}_a : \mu_1 < \mu_2$.

Exercise 3. Let $(X_1, Y_1), \dots, (X_n, Y_n) \stackrel{\text{i.i.d.}}{\sim} \text{BVN}(\mu_x, \mu_y, \sigma_x^2, \sigma_y^2, \rho)$, where $\mu_x, \mu_y, \sigma_x^2, \sigma_y^2, \rho$ are all unknown. Formulate tests for testing $\mathcal{H}_0 : \mu_x = \mu_y$ against

- (a) $\mathcal{H}_a : \mu_x \neq \mu_y$.
- (b) $\mathcal{H}'_a : \mu_x > \mu_y$.
- (c) $\mathcal{H}''_a : \mu_x < \mu_y$.

Exercise 4. Let $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Geometric}(p_1)$ and $Y_1, \dots, Y_m \stackrel{\text{i.i.d.}}{\sim} \text{Geometric}(p_2)$ be two independent samples. Formulate the likelihood ratio test, Wald's test and Rao's score test for testing $\mathcal{H}_0 : p_1 = p_2$ against $\mathcal{H}_a : p_1 \neq p_2$. How would you test $\mathcal{H}_0 : p_1 = p_2$ against $\mathcal{H}_a : p_1 > p_2$?

Exercise 5. For the following scenarios, find the coverage probabilities and the confidence coefficients of the confidence intervals/sets.

- (a) $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Bernoulli}(p)$. $I = I(\mathbf{X}) = [\bar{X} - 0.1, \bar{X} + 0.1]$.
- (b) $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Poisson}(\lambda)$. $I_1(\mathbf{X}) = [\bar{X} - 1, \bar{X} + 1]$, $I_2(\mathbf{X}) = [\bar{X}, \infty)$.
- (c) $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Normal}(\mu, 1)$. $I_1(\mathbf{X}) = [\bar{X} - 3, \bar{X} + 3]$, $I_2(\mathbf{X}) = [\bar{X}, \infty)$.
- (d) $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Normal}(\mu, \sigma^2)$. $I(\mathbf{X}) = [\bar{X} - 2, \bar{X} + 2]$.
- (e) $X \sim \text{Laplace}(\mu, 1)$. $I(X) = [X - 1, X + 2]$.

Exercise 6. Let $f_\theta(\cdot)$ be a pmf/pdf from the location family. Recall that this means $f_\theta(x) = f_0(x - \theta)$, where f_0 is completely known. In particular, for $X \sim f_\theta$, $X - \theta \sim f_0$. Let $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} f_\theta(\cdot)$.

- (a) Show that for any $a < b$, $\mathbb{P}_\theta(a < \bar{X} - \theta < b)$ is free of θ .
- (b) Deduce that the coverage probability of any confidence interval (CI) of the form $[\bar{X} - a, \bar{X} + b]$ is free of θ .
- (c) Use part (b) to show that a $100(1 - \alpha)\%$ CI for θ is $[\bar{X} - c_{\alpha/2}, \bar{X} + c_{\alpha/2}]$, where $c_{\alpha/2}$ satisfies $\mathbb{P}_0(|\bar{X}| \leq c_{\alpha/2}) = 1 - \alpha$. Such a $c_{\alpha/2}$ can be (in principle) obtained from the knowledge of f_0 only.

Exercise 7. Let $f_\theta(\cdot)$ be a pmf/pdf from the scale family, i.e., $f_\theta(x) = \frac{1}{\theta} f_1\left(\frac{x}{\theta}\right)$, where f_1 is completely known. In particular, for $X \sim f_\theta$, $\frac{X}{\theta} \sim f_1$. Let $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} f_\theta(\cdot)$.

- (a) Show that for any $a < b$, $\mathbb{P}_\theta(a < \bar{X}/\theta < b)$ is free of θ .
- (b) Deduce that the coverage probability of any CI of the form $[\bar{X}/a, \bar{X}/b]$ is free of θ .
- (c) Show that a $100(1 - \alpha)\%$ CI for θ is $[\bar{X}/c_{\alpha_1}, \bar{X}/c_{1-\alpha_2}]$, where c_α satisfies $\mathbb{P}_1(\bar{X} \geq c_\alpha) = 1 - \alpha$ and $\alpha_1 + \alpha_2 = \alpha$. Such c_{α_1} and $c_{1-\alpha_2}$ can (in principle) be obtained from the knowledge of f_1 only.
- (d) Show that parts (a)–(c) also hold if, instead of $\bar{X}$, we consider $X_{(n)}$.
- (e) Is it possible to use $X_{(n)}/X_{(1)}$ to construct a CI for θ ? Give justification.
- (f) Revisit Exercise 6. Is it possible to construct a CI for θ in that problem based on $X_{(n)} - X_{(1)}$? Justify.